

# Naive Bayes

CS60050: Machine Learning

Abir Das & Ayan Chaudhury

IIT Kharagpur

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# Agenda

- § Understand the philosophy of generative and discriminative schools of Machine Learning.
- § Learn about the “Naive Bayes” assumption
- § Employ the naivity assumption for discrete and continuous features

# Resources

- § Machine Learning: A Probabilistic Perspective by Kevin P. Murphy.
- § Andrew Ng's CS229 Lecture Notes

# Two Ways to Classify

We want to learn a mapping from features  $\mathbf{x}$  to class labels  $y \in \{0, 1\}$ .

## Discriminative (e.g., Logistic Reg)

- § Learn  $P(y|\mathbf{x})$  directly.
- § Tries to find the **decision boundary** separating classes.
- § "Given features  $\mathbf{x}$ , which class is it?"

## Generative (e.g., Naive Bayes)

- § Learn  $P(\mathbf{x}|y)$  and  $P(y)$ .
- § Models the **distribution** of data for each class separately.
- § "What does  $\mathbf{X}$  look like if it is a 'cat'?"

Prediction using Bayes Rule:

$$\hat{y} = \arg \max_y P(y|\mathbf{x}) = \arg \max_y \frac{P(\mathbf{x}|y)P(y)}{P(\mathbf{x})} = \arg \max_y \underbrace{P(\mathbf{x}|y)}_{\text{Likelihood}} \underbrace{P(y)}_{\text{Prior}}$$

## Motivation: The “Ideal” Generative Model

Ideally, we would just learn the full **Joint Distribution**  $P(x_1, \dots, x_D, y)$ . If we had this, we could answer ANY query (classification, missing data, etc.).

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§  $x_1$ : Fever (0/1)

§  $x_2$ : Cough (0/1)

§  $y$ : Flu (0/1)

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The Full Joint Distribution  $P(x_1, x_2, y)$  is a table of  $2^3 = 8$  entries:

| $x_1$ (Fever) | $x_2$ (Cough) | $y$ (Flu) | Prob |
|---------------|---------------|-----------|------|
| 0             | 0             | 0         | 0.40 |
| 0             | 0             | 1         | 0.01 |
| 0             | 1             | 0         | 0.05 |
| 0             | 1             | 1         | 0.05 |
| 1             | 0             | 0         | 0.05 |
| 1             | 0             | 1         | 0.10 |
| 1             | 1             | 0         | 0.04 |
| 1             | 1             | 1         | 0.30 |

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**We can answer ANY query!**

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| 1             | 1             | 0         | 0.04 |
| 1             | 1             | 1         | 0.30 |

§ **Prediction:**  $P(\text{Flu}|\text{Fever, Cough})?$

$$\frac{P(1,1,1)}{P(1,1,0)+P(1,1,1)} = \frac{0.30}{0.04+0.30} \approx 0.88$$

§ **Marginal:** Probability of Fever?

Sum rows where  $x_1 = 1$ :

$$0.05 + 0.10 + 0.04 + 0.30 = 0.49$$

§ **Missing Data:**  $P(\text{Flu}|\text{Cough})?$

Sum over Fever cases (0/1) and normalize.

# Missing Data

§  $x_1$ : Fever (0/1)

§  $x_2$ : Cough (0/1)

§  $y$ : Flu (0/1)

$$\begin{aligned} P(\text{Flu}|\text{Cough}) &= P(y = 1|x_2 = 1) = \frac{P(y = 1, x_2 = 1)}{P(x_2 = 1)} \\ &= \frac{\sum_{x_1} P(y = 1, x_1, x_2 = 1)}{\sum_y \sum_{x_1} P(y, x_1, x_2 = 1)} \\ &= \frac{0.05 + 0.30}{0.05 + 0.05 + 0.04 + 0.30} = \frac{0.35}{0.44} \\ &= 0.795 \end{aligned}$$

## Takeaway

Even though we didn't measure Fever ( $x_1$ ), we could still use the Joint Distribution to provide an exact probability. This is the power of Generative Models!

## Example 2: Image Classification

Now imagine a tiny  $30 \times 30$  binary image ( $D = 900$  pixels).



$30 \times 30$  image

### The Joint Distribution Table:

- § Number of rows =  $2^{900} \approx 10^{270}$
- § Assuming 1 row takes 1 nanosecond to read...
- § It would take billions of years just to scan the table!

**Prediction:** 
$$P(Y = 5 | X_1, X_2, \dots, X_{900}) = \frac{P(X_1, X_2, \dots, X_{900} | Y=5)P(Y=5)}{P(X_1, X_2, \dots, X_{900})}$$

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### Conclusion

For real-world problems (Text, Images, Genetics), we **cannot** estimate the full joint distribution table. We neither have enough data nor enough time to fill and read  $10^{270}$  rows.

**We need a simplifying assumption! → Naive Bayes.**

# The “Naive” Solution

To make learning feasible, we make a strong assumption to reduce parameters.

**Naive Bayes Assumption:** The features  $x_1, \dots, x_D$  are **conditionally independent** given the class  $y$ .

$$P(\mathbf{x}|y) = P(x_1, x_2, \dots, x_D|y) = \prod_{j=1}^D P(x_j|y)$$

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## The Parameter Savings:

- § Instead of one giant table of size  $2^D$ , we learn  $D$  small tables.
- § For binary features: we need just 1 parameter per feature per class.
- § Total parameters:  $2D + 1$  (Linear  $O(D)$  complexity instead of Exponential  $O(2^D)$ !).

# The Naive Bayes Classifier

## § Given:

- ▶ Prior  $P(y)$
- ▶  $D$  conditionally independent features  $x_1, \dots, x_D$ , given the class  $y$
- ▶ For each feature, we specify  $P(x_j|y)$

## § Classification decision rule:

- ▶  $y^* = \arg \max_y P(y)P(x_1, \dots, x_D|y) = \arg \max_y P(y) \prod_j P(x_j|y)$

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## § Let us take an example involving two classes. In case of two classes, $y \in \{0, 1\}$ , we predict that $y = 1$ if:

$$\frac{P(y = 1) \prod_j P(x_j|y = 1)}{P(y = 0) \prod_j P(x_j|y = 0)} > 1 \quad (1)$$

# Naive Bayes: Two Classes

§ Assuming Boolean features  $x_j \in \{0, 1\}$ , let

$$p_j = P(x_j = 1|y = 1), \text{ then } 1 - p_j = P(x_j = 0|y = 1)$$

§ Hence:  $P(x_j|y = 1) = p_j^{x_j} (1 - p_j)^{(1-x_j)}$

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§ Similarly,

$$q_j = P(x_j = 1|y = 0), \text{ then } 1 - q_j = P(x_j = 0|y = 0)$$

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§ Then eqn. (1) implies,

$$\frac{P(y = 1) \prod_j p_j^{x_j} (1 - p_j)^{(1-x_j)}}{P(y = 0) \prod_j q_j^{x_j} (1 - q_j)^{(1-x_j)}} = \frac{P(y = 1) \prod_j \left(\frac{p_j}{1-p_j}\right)^{x_j} (1 - p_j)}{P(y = 0) \prod_j \left(\frac{q_j}{1-q_j}\right)^{x_j} (1 - q_j)} > 1$$
$$\frac{P(y = 1)}{P(y = 0)} \prod_j \left(\frac{\frac{p_j}{1-p_j}}{\frac{q_j}{1-q_j}}\right)^{x_j} \prod_j \left(\frac{1-p_j}{1-q_j}\right) > 1 \quad (2)$$

# Naive Bayes: Two Classes

§ Take logarithm; we predict  $y = 1$ , if:

$$\log \frac{P(y=1)}{P(y=0)} + \underbrace{\sum_j \log \frac{1-p_j}{1-q_j}}_{b_j} + \sum_j \underbrace{\left( \log \frac{p_j}{1-p_j} - \log \frac{q_j}{1-q_j} \right)}_{\theta_j} x_j > 0 \quad (3)$$

# Naive Bayes: Learning Parameters

- § How do we estimate the probabilities from training data?
- § We use **Maximum Likelihood Estimation (MLE)**, which boils down to simple counting.

▶ **Prior Probabilities**  $P(Y)$ :

$$\hat{P}(Y = 1) = \frac{\# \text{ examples with } Y = 1}{\text{Total } \# \text{ examples}}$$

▶ **Conditional Probabilities**  $P(X_i = 1|Y = y)$ :

$$\hat{P}(X_i = 1|Y = 1) = \frac{\# \text{ examples with } Y = 1 \text{ AND } X_i = 1}{\# \text{ examples with } Y = 1}$$

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- § Similarly for  $P(X_i = 1|Y = 0)$ .
- § Since features are binary,  $P(X_i = 0|Y) = 1 - P(X_i = 1|Y)$ .

# The Dataset: Will I Play Tennis Today?

| Day | Sky   | Temp | Humid  | Play? |
|-----|-------|------|--------|-------|
| D1  | Sunny | Warm | Normal | Yes   |
| D2  | Rainy | Warm | High   | Yes   |
| D3  | Sunny | Cold | Normal | Yes   |
| D4  | Rainy | Cold | High   | No    |
| D5  | Sunny | Warm | High   | No    |

§ **Training Data** ( $N = 5$ ):

§ **Test Instance:**

$x = (\text{Sunny}, \text{Warm}, \text{High})$

§ **Goal:** Predict Play? (Yes/No)

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**Class: Yes** ( $N_{yes} = 3$ )

§  $P(\text{Yes}) = 3/5$

§  $P(\text{Sunny}|\text{Yes}) = 2/3$

§  $P(\text{Warm}|\text{Yes}) = 2/3$

§  $P(\text{High}|\text{Yes}) = 1/3$

$$\text{Score}_{Yes} = \frac{3}{5} \times \frac{2}{3} \times \frac{2}{3} \times \frac{1}{3} = 0.089$$

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**Class: No** ( $N_{no} = 2$ )

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§  $P(\text{Warm}|\text{No}) = 1/2$

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$$\text{Score}_{No} = \frac{2}{5} \times \frac{1}{2} \times \frac{1}{2} \times 1 = 0.1$$

**Prediction:**  $0.1 > 0.089 \implies \text{No}$

## Scenario 2: Slight Data Change... Big Problem!

§ Suppose, in our “Yes” examples ( $N = 3$ ), the “Temp” column always sees **Cold** days.

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Calculation for  $P(\text{Yes}|x)$

$$\begin{aligned} P(\text{Yes}) \times P(\text{Sunny}|\text{Yes}) \times P(\text{Warm}|\text{Yes}) \times \dots \\ = \frac{3}{5} \times \frac{2}{3} \times \frac{0}{3} \times \dots = 0 \end{aligned}$$

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§ **The “Zero Frequency” Problem:** The probability is zero just because we haven’t seen a “Warm” day for “Play=Yes” yet. This vetoes all other strong evidence (like Sunny!).

# The Fix: Laplace Smoothing

§ Are you thinking “such a complicated term this is!! LAPLACE SMOOTHING!”

§ **Idea:** Add a “virtual count” of 1 to every value.

$$\hat{P}(x_i|y) = \frac{\text{Count}(x_i, y) + 1}{\text{Count}(y) + |V|}$$

where  $|V|$  is the “**number**” of values that the feature can take.

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where  $|V|$  is the “**number**” of values that the feature can take.

**Re-calculating**  $P(\text{Warm}|\text{Yes})$ :

- ▶  $\text{Count}(\text{Warm}, \text{Yes}) = 0$
- ▶  $\text{Count}(\text{Yes}) = 3$
- ▶ Possible Temps ( $|V|$ ) = 2 {Warm, Cold}

$$\hat{P}(\text{Warm}|\text{Yes}) = \frac{0 + 1}{3 + 2} = \frac{1}{5} = \mathbf{0.2}$$

*Now the probability is small, but non-zero!*

# Handling Continuous Features

- § So far, we have discussed Naive Bayes with discrete features (e.g., word counts, sky condition).
- § **Question:** What if our features  $\mathbf{x} = (x_1, \dots, x_d)$  are **continuous** real numbers?
  - ▶ Example: Classifying if a person is "Healthy" or "Sick".
  - ▶ Features: Height (cm), Weight (kg), Temperature ( $^{\circ}\text{F}$ ).
- § We cannot use simple counting/tables because the probability of observing an exact real number (e.g., height = 170.0001 cm) is zero.

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- § We cannot use simple counting/tables because the probability of observing an exact real number (e.g., height = 170.0001 cm) is zero.
- § **Solution:** We need a **probability density function (PDF)** to model  $P(x_j|Y)$ . The most common choice is the **Gaussian (Normal) Distribution**.

# Gaussian Naive Bayes Assumption

We assume that for each class  $y \in \{0, 1\}$ , the continuous features  $x_i$  are distributed according to a Gaussian distribution.

## Model Assumptions

§ **Class Prior:**  $Y \sim \text{Bernoulli}(\phi)$

§ **Conditional Distributions:**

$$P(x_j|Y = y) = \frac{1}{\sqrt{2\pi\sigma_{jy}^2}} \exp\left(-\frac{(x_j - \mu_{jy})^2}{2\sigma_{jy}^2}\right)$$

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This means for each feature  $j$  and each class  $y$ , we need to estimate two parameters:

§ Mean  $\mu_{jy}$ : The average value of feature  $j$  for class  $y$ .

§ Variance  $\sigma_{jy}^2$ : How spread out feature  $j$  is for class  $y$ .

Imagine we have one feature  $x$  (e.g., Temperature) and two classes ( $Y = 0$  Healthy,  $Y = 1$  Sick).

Figure 1 shows a plot of the joint density function of temperature and the presence of a cold. The x-axis represents Temperature in degrees Celsius, ranging from 35 to 42. The y-axis represents Density, ranging from 0 to 0.6. There are two probability density functions plotted:  $P(x|Y=0)$  (blue curve) and  $P(x|Y=1)$  (red curve). The blue curve is centered around 36.5°C, and the red curve is centered around 38.5°C. The area under the blue curve is shaded light blue, and the area under the red curve is shaded light red.

# Learning Parameters (MLE)

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- § Given dataset  $\{(x^{(1)}, y^{(1)}), \dots, (x^{(N)}, y^{(N)})\}$ :

▶ **Means:**

$$\hat{\mu}_{jy} = \frac{\sum_{i=1}^N \mathbb{I}(y^{(i)} = y) \cdot x_j^{(i)}}{\sum_{i=1}^N \mathbb{I}(y^{(i)} = y)}$$

(Average of feature  $j$  for all examples where class is  $y$ )

▶ **Variances:**

$$\hat{\sigma}_{jy}^2 = \frac{\sum_{i=1}^N \mathbb{I}(y^{(i)} = y) \cdot (x_j^{(i)} - \hat{\mu}_{jy})^2}{\sum_{i=1}^N \mathbb{I}(y^{(i)} = y)}$$

(Variance of feature  $j$  for all examples where class is  $y$ )

# Concrete Example

**Dataset (Height in feet):**

§ Class 0: {5.0, 5.5, 6.0}

§ Class 1: {6.0, 6.2}

# Concrete Example

## Dataset (Height in feet):

§ Class 0: {5.0, 5.5, 6.0}

§ Class 1: {6.0, 6.2}

## Step 1: Estimate Means

$$§ \hat{\mu}_0 = \frac{5.0+5.5+6.0}{3} = 5.5$$

$$§ \hat{\mu}_1 = \frac{6.0+6.2}{2} = 6.1$$

# Concrete Example

**Dataset (Height in feet):**

§ Class 0: {5.0, 5.5, 6.0}

§ Class 1: {6.0, 6.2}

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**Step 2: Estimate Variances**

$$§ \hat{\sigma}_0^2 = \frac{(5.0-5.5)^2 + (5.5-5.5)^2 + (6.0-5.5)^2}{3} = \frac{0.25+0+0.25}{3} = 0.167$$

$$§ \hat{\sigma}_1^2 = \frac{(6.0-6.1)^2 + (6.2-6.1)^2}{2} = \frac{0.01+0.01}{2} = 0.01$$

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**Prediction for new  $x = 5.8$ :** Compute  $P(x = 5.8|Y = 0)P(Y = 0)$  vs  $P(x = 5.8|Y = 1)P(Y = 1)$  using Gaussian formula.

$$P(x = 5.8|Y = 0) = \frac{1}{\sqrt{2\pi(0.167)}} e^{-\frac{(5.8-5.5)^2}{2(0.167)}} \approx 0.98 \times e^{-0.27} \approx \mathbf{0.75}$$

$$P(x = 5.8|Y = 1) = \frac{1}{\sqrt{2\pi(0.01)}} e^{-\frac{(5.8-6.1)^2}{2(0.01)}} \approx 3.99 \times e^{-4.5} \approx \mathbf{0.04}$$

(Ignoring priors for simplicity, Class 0 is much more likely)

# The “Linear” Connection

- § **Question:** What does the decision boundary look like?
- § If we assume the variance is **shared** across classes ( $\sigma_{j0}^2 = \sigma_{j1}^2 = \sigma^2$ ), something magical happens.

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$$\log \frac{P(Y = 1|x)}{P(Y = 0|x)} = \log \frac{P(Y = 1)}{P(Y = 0)} + \sum_j \log \frac{P(x_j|Y = 1)}{P(x_j|Y = 0)}$$

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- § Substituting the Gaussian formula:

$$\log \frac{\exp(-(x_j - \mu_{j1})^2/2\sigma^2)}{\exp(-(x_j - \mu_{j0})^2/2\sigma^2)} = \frac{-(x_j - \mu_{j1})^2 + (x_j - \mu_{j0})^2}{2\sigma^2}$$

- § The quadratic terms  $x_j^2$  cancel out! We are left with terms linear in  $x_j$ .

# The “Logistic Regression” Connection

Since the quadratic terms cancel (under shared variance assumption), the log-odds is a linear function:

$$\log \frac{P(Y = 1|\mathbf{x})}{P(Y = 0|\mathbf{x})} = \boldsymbol{\theta}^T \mathbf{x} + b$$

This implies the posterior probability is a **Sigmoid**:

$$P(Y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(\boldsymbol{\theta}^T \mathbf{x} + b)}}$$

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This specific model—Naive Bayes with continuous features assumed to be Gaussian—is formally known as **Gaussian Discriminant Analysis (GDA)**.

## Key Takeaway

Gaussian Naive Bayes (with shared variance) is a **Generative** classifier that produces a **Linear** decision boundary, exactly like Logistic Regression!

§ GDA: Learns  $P(X|Y)$  (Means/Variiances)  $\rightarrow$  gets  $\boldsymbol{\theta}$ .

§ Logistic Regression: Learns  $\boldsymbol{\theta}$  directly.

# Summary: Naive Bayes & GDA

- § **Generative Approach:** We model  $P(\mathbf{x}|y)$  and  $P(y)$  to estimate the joint distribution  $P(\mathbf{x}, y)$ , then use Bayes Rule for classification.
- § **Discrete Features:** Use **Naive Bayes** (Bernoulli/Multinomial) with Laplace Smoothing to handle zero probabilities.
- § **Continuous Features:** Use **Gaussian Discriminant Analysis (GDA)**. We estimate Means and Variances.
- § **Key Insight:** If we assume shared covariance in GDA, the decision boundary is linear, and the posterior form is identical to **Logistic Regression**.
  - ▶ **GDA** is more data-efficient (lower variance) if the Gaussian assumption is true.
  - ▶ **Logistic Regression** is more robust (lower bias) if the assumption is wrong.
- § GDA with shared covariance is often called **Linear Discriminant Analysis (LDA)**. Allowing different covariances, makes it **Quadratic Discriminant Analysis (QDA)**.

Next Class: Bias and Variance